

8/17

Review of UFT in $\mathbb{Z} = \{0, \pm 1, \pm 2, \dots\}$

Def (Divisibility): Let $a, b \in \mathbb{Z}$ with $a \neq 0$. We say that

a divides b / a is a divisor of b / b is divisible by a ,

if there is $c \in \mathbb{Z}$ s.t. $b = ac$, namely, $\frac{b}{a} = c \in \mathbb{Z}$.

E.g. 1. $3 \mid 6$

2. $4 \mid -12$

Properties:

1. Reflexivity: $a \mid a$ ($a \neq 0$).

2. Anti-symmetry: $a \mid b$, $b \mid a \Rightarrow b = \pm a$.

3. Transitivity: $a \mid b$, $b \mid c \Rightarrow a \mid c$.

4. Linearity: $a|b, a|c \Rightarrow a|xb+yc, x, y \in \mathbb{Z}$.

5. Size: $a|b (b \neq 0) \Rightarrow |a| \leq |b|$.

Defn (Unit). Let R be a comm ring (with 1). A unit

in R is an element $r \in R$ s.t. $rs = sr = 1$ for some $s \in R$.

Q: Units in $\mathbb{Z}$?

A: ± 1 .

Defn (Prime & Composite) $n > 1$ is called a prime number

if it has only 1 and itself as positive divisors.

Otherwise, n is called a composite number.

Lem 1.1. Every $n > 1$ has a prime divisor.

Pf. ① $n = \text{prime}$ $\checkmark$.

② $n = \text{composite}$: pick the smallest positive divisor

$$1 < p \leq n.$$

Claim: p is prime.

If $d \mid p$ - then $d \mid n$. By minimality p and

$d \leq p$ - we see $d = 1$ or $d = p$. So p is prime. $\square$

Thm (Fundamental Thm of Arithmetic) Every $n > 1$ can be

written as a product primes, uniquely up to the order

of the prime factors.

E.g. $60 = 2 \cdot 2 \cdot 3 \cdot 5 = 2 \cdot 3 \cdot 2 \cdot 5 = 5 \cdot 3 \cdot 2 \cdot 2 = \dots$

Thm (Division Algorithm). Let $a, b \in \mathbb{Z}$, $b \neq 0$. Then there are

unique $q, r \in \mathbb{Z}$ with $0 \leq r < |b|$, s.t.

$$a = q \cdot b + r.$$

↑ ↑ ↑
dividend quotient remainder.